Chapter 13

Drug Information Training for Pharmacists

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Learning Objectives:

Objective 13.1 Explain why pharmacists in any work setting require well-founded drug information skills.
Objective 13.2 Review drug information training strategies for pharmacy students.
Objective 13.3 Categorize drug information skills to be covered during training.
Objective 13.4 Plan a week-wise follow-up of learning objectives achieved in drug information training.

OBJECTIVE 13.1. EXPLAIN WHY PHARMACISTS IN ANY WORK SETTING REQUIRE WELL-FOUNDED DRUG INFORMATION SKILLS

Drug information is a service provided by pharmacists in their daily practice. Irrespective of the work settings, community, hospital, consulting, and clinical pharmacists provide drug information services. The level of drug information provided may vary, however. For example, community pharmacists may answer more simple drug-specific questions of their patients (e.g., dosing, generic availability). Hospital pharmacists may answer drug-specific questions (e.g., intravenous administration/compatibility), and pharmacists exclusive clinical practice may answer more patient-specific drug information questions (e.g., alternative therapy due to patient allergy). Drug information pharmacists who have specialized in operating and sustaining a drug information service may tend to answer more complex questions that need a thorough review of the literature. Answering these in-depth, complex questions requires additional time to assess and evaluate patient-specific information and conduct appraisal and analysis of the appropriate literature. The type of drug information training for such positions varies accordingly. To become a clinical specialist in a drug information center (e.g., in a challenging hospital environment), advanced residency training shall be required and may take one to 2 years. Pharmacists, who work as full-time drug/poison information specialists regionally (i.e., not restricted to a hospital), are also commonly contacted by healthcare professionals and the general public to answer drug information questions with varying degrees of complexity,[1] whereas others work for the pharmaceutical industry, health insurance companies, pharmacy benefit management companies, and governmental agencies. Some countries restrict drug information services to the public, determining that incomplete medical information without professional consultation might increase inappropriate self-medication. A consultation with pharmacists shall result in a proper understanding of the medication information.[2]

Pharmacists, being one of the most accessible healthcare professionals, are often approached by the public for their drug information needs. Pharmacists may also contribute to the management of minor ailments, guiding the use of over-the-counter (OTC) drugs and herbal products. They should also be prepared to counsel safe drug use to pregnant women, as well as children and the elderly.[3,4] According to the Consumer Healthcare Products Association website, 93% of US adults prefer to treat their minor ailments with OTC drugs before seeking professional care, and 85% of US parents do the same with their children.[5] OTC drugs allow consumers more accessibility. However, their use is not free from risk.

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Drug interactions, cross-reactivity among drugs, toxicity, dosing errors, and continuation of therapy should be considered when counseling patients about their OTC medication use. The provision of accurate and helpful drug information by pharmacists is important, as consumers often forget to discuss their OTC drug use with other healthcare professionals. Identification and appropriate use of references ideally suited for these patient care issues are essential for the busy pharmacist to guarantee proper care of these patients with efficient use of time and resources.

Special attention has also been given to off-label drug use because off-label drug use has been a common practice but often is not supported by strong evidence.[6] Prescribers need to be informed about the availability of drugs for certain indications, as well as details of appropriate dosage regimens. Accepting information from the pharmaceutical company by prescribers shall affect prescribing quality.[7] Unbiased information provided by pharmacists will improve the practice. Informed practice leads to safer and cost-effective drugs use. Information about affordable treatment options reduces healthcare expenses. Consumers and nurses also need reliable information on proper administrations of drugs and how to monitor for adverse events. Drug information training helps pharmacists to promote rational drug use, decrease medication errors, and alleviate adverse effects.

In this era of improved collaborative practice, healthcare professionals share their knowledge and skills across services for the benefit of better patient outcomes. This multidisciplinary approach to patient care increases interactions between other healthcare professionals (e.g., doctors, nurses, dentists) and pharmacists resulting in increased drug information needs. All pharmacists should possess the basic drug information skills to meet the demand of their specific practice.[8]

Drug information skills and general pharmacotherapy knowledge are also useful on an administrative level. Drug information and hospital pharmacists take part in the development and maintenance of the hospital formulary system. Recommendations to include or remove a drug from the formulary need to be supported with sufficient drug information from the literature. The same skills are useful for the development of clinical guidelines within an institution, prior authorization programs, and other drug management programs. Quality assessment of how medications are being used can result in the development of treatment protocols or other guided use strategies for the medications evaluated. This medication use evaluation (MUE) utilizes current knowledge about pharmacotherapy and an assessment of how the medication is being used.

Participating in healthcare research requires drug information skills. Drug information skills equip professionals to conduct better literature searches and evaluations. Pharmacists with enhanced literature searching skills are in a better position to collect comprehensive information from the Internet, databases, and various article-retrieval systems. Critical literature appraisal skills needed to properly analyze and determine if a citation should be included in a research project. Pharmacists skilled in writing publications for various journal articles may contribute to a research team or project by better understanding the submission process and/or writing style for a particular journal. Because of a combination of healthcare and drug-specific knowledge, search and critical appraisal skills, and the writing and online journal article submission expertise, pharmacists are often preferred members of healthcare research teams. Literature evaluation and management skills help to elevate overall evidence-based practice and research.

OBJECTIVE 13.2. REVIEW DRUG INFORMATION TRAINING STRATEGIES FOR PHARMACY STUDENTS

Education starts with information management. Students are taught how to convert received information into knowledge. It is up to them to figure out how to convert that knowledge into wisdom. The abundance of drug information resources available and varying accessibility makes it difficult to manage information without technological skills.[9] Drug information training helps students quickly find relevant information and convert it into an applicable form in clinical practice.[1]

Pharmacy students undergo didactic and experiential education in drug information management. Didactic education is commonly delivered as lectures, interactive learning situations, and structured activities. Experiential education is generally delivered in Advanced Pharmacy Practice Experience (APPE); it is called an internship or clerkship. After graduation, drug information residencies are available to master other skills associated with more advanced drug information-related activities (e.g., formulary management, personnel management, policy/procedure creation, and teaching). Based on the level and type of education program, specific learning objectives need to be developed.

Skills used to provide drug information are taught classically by using the modified systematic approach to answering queries.[10,11] Additional training modalities include journal club discussions, drug profile presentations, therapeutic management presentations, research review, practice research, publishing, and interactions with healthcare professionals or the public. The objectives of these training programs are to ultimately provide unbiased, quality information in a reasonable time frame in an understandable format for the person making the inquiry.

The intentional and unintentional bias of the researchers/authors may result in unreliable articles. With a heavy influence of pharmaceutical marketing, bias within the literature can and does happen.[12] Critical appraisal skills compliment knowledge and attitudes to identify methodological bias that could be associated with conflicts of interests of the authors and the quality of the studies and literature.[13] These literature evaluation skills are required to review the applicability of information in clinical practice.

Other biases regarding the retrieval or dissemination of information may exist. For example, individual pharmacists sometimes heavily depend on sources of information that they are comfortable with and may miss vital information available in other resources. In doing so, pharmacists may unknowingly provide inadequate or wrong information to other healthcare professionals.[14] Yet, if the information being provided has been verified and validated in tertiary, secondary, and primary references, pharmacists should not accept conformity bias to avoid providing clinicians when information is incorrect, biased, or outdated.[15] Healthcare professionals are especially unwilling to expose their lack of understanding on some medical topics.[16] Therefore, confidentiality and good communication skills are important.

The scientific quality of the information being delivered can be ensured by observance and implementation of evidence-based practice (EBP). Drug information centers can and often work as a focal point in the EBP of a hospital. Healthcare professionals need to incorporate data from the literature, as well as their own research. This should be incorporated with experiences from the practice site along with patient observations and values.[17] Drug information pharmacists work to retrieve relevant articles and other up-to-date information at the request of various healthcare professionals within an institution. Additionally, they can contribute to the development and maintenance of evidence-based clinical guidelines and the hospital formulary system to improve medication use practices and overall patient care.[1]

Quality assurance is essential to systematically plan and implement drug information education and training in pharmacy programs. Constructive feedback is used to reassure the student about the skills evaluated and requirements of further training to achieve any unmet learning objectives. The training needs to be comprehensive for achieving these skills, as well as the associated knowledge and attitudes to practice drug information in a variety of settings. The accomplishment of these outcomes also assures that the trainee is self-aware of the need to keep abreast of the current standards in practice.

OBJECTIVE 13.3. CATEGORIZE DRUG INFORMATION SKILLS TO BE COVERED DURING TRAINING

Drug information skills are best improved and attained by using a systematic approach to answer questions. All of the seven steps listed below assess combinations of both general (e.g., communication, clinical, research) and specific (e.g., search, appraisal, writing) identifiable skills.[18]

1. **Secure demographics of the requestor.** It is important to know who is asking the question and who wants to know the answer (not always the same person). The type of person you are talking to may dictate where you will go to find the answer. For example, the information context and level may vary between different health professionals and patients. However, a physician may be more interested in the latest scientific advancement from the primary literature, whereas a patient may require knowing the usual pattern of drug use at different level of understanding.
   - **a.** Requires:
     - **i.** Communication skills.
     - **ii.** Associated attitudes for developing and maintaining trust of requesters.
   - **b.** Information to be gathered:
     - **i.** Requestor's name.
     - **ii.** Requestor's location and mobile number.
     - **iii.** Requestor's affiliation (institution or practice), if a healthcare professional.
     - **iv.** Requestor's frame of reference (status—type of profession, nurse, physician, pharmacist, student, etc.).
     - **v.** The best way to contact the requestor to provide the response.
2. **Obtain pertinent background information.** It is important to obtain this information at the time of the initial contact because it is sometimes difficult to get back in touch with a busy health professional.
   - **a.** Requires
     - **i.** Communication skills.
     - **ii.** Associated knowledge about pharmacotherapy and medical practices in the work setting.
     - **iii.** Associated attitudes for confidentiality of information and patient benefit.

b. Information to be gathered:
   i. Resources the requestor has already checked—to avoid duplication of effort or validate what the requestor has already read to see what might be missing, misunderstood, or if the information may be outdated.
   ii. Patient-specific information: medical problems, prescription, OTC, and herbal medications, renal function, hepatic function, and other pertinent information.
   iii. Urgency of the request—the time frame for the response.

3. **Determine and categorize ultimate question.** It is important to understand what the requestor wants. The way they phrased the question may give you a clue to what information the person really needs, so listen carefully and then ask good questions of the requestor to identify what really needs to be answered.
   a. Requires
      i. Communication skills.
      ii. Associated knowledge about therapeutics and patient conditions, intervention, comparable therapeutic options, positive and negative outcomes expected (Patient Intervention Comparator Outcome—PICO).
      iii. Associated skills for helping requestors to develop an answerable question while limiting exposure for lack of understanding.
   b. Tips as you gather this information:
      i. On the form where you record your information, have a question background section where you make notes about what the requestor wants.
      ii. Only when you understand what information is needed should you record the question you intend to answer.
      iii. Verify with the requestor that this is indeed the question that you are trying to answer.
      iv. Categorize this question.
         (1). There are numerous examples, but it might be as simple as one of these: adverse drug reaction/contraindication, availability, dosage, drug compatibility/stability, drug interaction, therapeutic use, identification, formulation, or toxicity.
         (2). It is good practice to develop standard templates for appropriate question prompts for specific categories, e.g. for an ADR record dechallenge and rechallenge information and time course of the reaction, etc.
4. **Develop searching strategy and conduct searches.** The time frame for the search, the type of professional requesting the information, and the classification of the question are key to deciding where to search the primary or the tertiary literature and which resource is preferred.
   a. Requires
      i. Literature searching skills.
      ii. Associated knowledge about primary, secondary, and tertiary literature sources of information.
      iii. Range of information available on the internet.
      iv. Associated attitudes for unbiased, quality, and comprehensive search.
5. **Perform evaluation, analysis, and synthesis of collected information.** The information you retrieve must be critically analyzed. Often there is more information obtained that can be easily communicated to the requestor, so you will need to refine what you intend to communicate.
   a. Requires
      i. Critical literature appraisal skills, scientific review skills, and application of knowledge.
      ii. Associated knowledge in clinical practice, especially therapeutics.
      iii. Associated attitudes for unbiased and scientific interpretation of information.
6. **Formulate and provide a response.** Often there is not a clear-cut answer. Therefore, it will be necessary to present the options to the requestor, then make your assessment and emphasize your recommendation in a way that will allow the requestor to make their own decision based on the evidence provided to them in your formal response.
   a. Requires
      i. Medical writing skills, communication skills.
      ii. Associated knowledge regarding background information specific to the question and general pharmacotherapy/scientific knowledge.
      iii. Associated attitudes for tailoring the response to the needs of the requestor.
   b. Research has been done on what a requestor is looking for in your response.[19] They want the answer to be:
      i. Timely and current
      ii. Pragmatic - clinically useful and meaningful and patient focused (these are covered in ix, x and xi)
      iii. Accurate
      iv. Complete, yet concise

v. Well referenced
vi. Clear and logical
vii. Objective and balanced
viii. Free of bias and flaws
ix. Applicable and appropriate for the specific circumstances
x. It anticipates and answers important related questions
xi. Lastly, it addresses management of patients or situations, if applicable.

7. **Conduct follow-up and documentation.** Documentation of the information provided is very important. Equally important, but often overlooked, there should be a follow-up to document how the recommendation was utilized and the outcome of this intervention.
   a. Requires:
      i. Communication and documentation skills.
      ii. Associated knowledge of relevant policies and practices in the documentation.
      iii. Associated attitudes for client-centered practice, self-development, and adhering to relevant policies and procedures.

In addition to answering queries, key skills to develop and evaluate during training are as follows:

- Critical literature appraisal through journal club review and evaluating news stories
- The ability to asses the benefits and potential harms of treatment and communicate this effectively
- Medical writing skills through writing and editing of drug monographs or other write-ups
- Interpretation skills of clinical practice guidelines and implementation within the practice
- Research skills through participating in practice research
- Communication skills through short educational presentations, reports, and publishing (which could be a newsletter or even a blog).

OBJECTIVE 13.4. PLAN A WEEK-WISE FOLLOW-UP OF LEARNING OBJECTIVES ACHIEVED IN DRUG INFORMATION TRAINING

In academic programs, the curriculum is commonly designed in three steps to provide mastery of learning drug information. (1) In didactic sessions, students learn the basics of references that are available to use and why certain situations would necessitate the use of one over the other. (2) Introductory Pharmacy Practice Experiences allow learners to observe drug information practices in different healthcare settings. (3) APPE drug information rotations are advanced experiences in which students practice drug information under supervision in a real setting, usually in a drug information center that serves a hospital.[20] Pharmacy graduates from other programs without such APPE drug information rotations are usually offered on-the-job training by an experienced pharmacist for the development of overall drug information skills. The following quality checks can be made to assure the development of relevant skills and competencies in drug information practices during the training of pharmacy undergraduates or pharmacists. Scales are used to assure each task is completed satisfactorily. A common example of an evaluation scale is as follows:

ACH—Achieved
SP—Satisfactory progress
NI—Needs improvement
N/A—Not applicable.

Feedback is given on completion of assigned tasks and usually at the end of a rotation. Recommendations are given for any additional tasks if required. A typical 5-week training experience for a hospital-based experience is illustrated in the following tables (Tables 13.1−13.4). Similar types of schedules and objectives can develop each type of learning situations (e.g., pharmacists' practice, drug information center, pharmacy benefits management company, or health insurance company); the key is to provide the student with a guided experience. In these examples each learning objective is followed by its action plans with multiple tasks.

Each task given to the trainee is based on planned objectives. Achievement of each learning outcome should be rated to assure minimum competency for practicing in a hospital setting. A simpler version of this program shall likely be sufficient in community practice.[21] This training also empowers the trainees with the skills for self-learning. With conscious practice and increased experiences, drug information skills will improve over time. Further training (i.e., residencies) shall be required for trainees/pharmacists to work as drug information specialists.

TABLE 13.1 Learning Objectives for Week 1

No	Topic: Drug Information Needs, Resources, and Evidence-Based Practice	
1	Objective 1: Review pharmacy and therapeutic committee (PTC)/formulary management system in the hospital.	
	Action Plans	Review constitution and function of the hospital PTC.
		Interview members of the hospital PTC and clinicians to assess their drug information needs.
		Review hospital formulary list and compare with others (e.g., National, WHO).
		Observe the formulary system (e.g., drug inventory, distribution, dispensing, administration and how the hospital manages potential and actual drug-related problems).
		Present/report about Objective 1.
2	Objective 2: Distinguish between evidence-based and empirical treatment.	
	Action Plans	Review evidence-based practice principles.
		Review evidence-based treatment guidelines within the hospital.
		Explain subjective and objective patient outcome information from clinical case studies reviewed.
		Identify evidence-based and empirical drug use in cases reviewed.
		Present/report about Objective 2.
3	Objective 3: Organize and appraise the quality of drug information sources.	
	Action Plans	Identify quality metrics (e.g., impact factor) in biomedical journals.
		Appraise the quality of information in drug information databases (e.g., Micromedex, Drugs.com, DailyMed).
		Identify search functions available in indexing services (e.g., PubMed, Google Scholar).
		Organize drug information resources available for efficient retrieval of information.
		Review and present modified systematic approach to answering questions.

TABLE 13.2 Learning Objectives for Week 2

No	Topic: Drug Information Service and Literature Evaluation	
1	Objective 4: Answer drug-specific queries by modified systematic approach (continued daily until completion of rotation).	
	Action Plans	Answer drug-specific queries from healthcare professionals using a primary source of information (e.g., original research, journal article).
		Validate information provided using a tertiary source (database/textbook).
		Analyze information and compile into a complete response with proper documentation.
		Respond to feedback from the requestor.
		Discuss in a small group about the drug information experience and strategies to improve the service.
2	Objective 5: Demonstrate responsibility in providing unbiased, quality drug information in a timely manner.	
	Action Plans	Review timeliness of providing drug information service.
		Review conflicts of interest in the resources used.
		Comply with hospital and national policies, as well as regulatory and ethical standards in providing a drug information service.
		Appraise the quality of sources of information used.
		Communicate effectively to healthcare professionals utilizing unbiased, quality drug information as part of evidence-based practice.

Continued

TABLE 13.2 Learning Objectives for Week 2—cont'd

No	Topic: Drug Information Service and Literature Evaluation	
3	Objective 6: Perform critical appraisal of a selected therapy article in journal club that is repeated possibly once per week. (See Appendix 1 for a rubric to assess the journal club presentation).	
	Action Plans	Select a therapy article for critical evaluation.
		Discuss methods of critical evaluation.
		Present and critically evaluate the selected article in a journal club.
		List strengths and weaknesses of the article.
		Justify how the findings from the study are applicable in the local healthcare practice.

TABLE 13.3 Learning Objectives for Week 3

No	Topic: Drug Information Service, Practice Research, and Pharmacotherapy	
1	Objective 7: Conduct retrospective practice research (e.g., drug utilization review, pharmacoeconomics, or pharmacovigilance) or prepare a manuscript for drug information/institution newsletter.	
	Action Plans	Collaborate with other healthcare professionals on planning practice research. Retrospective research from the specified hospital information system is preferred. Otherwise, anarrative review published in an institution's newsletter is acceptable.
		Perform the research to identify issues in healthcare delivery.
		Review literature for similar practice research or topic selected.
		Discuss with different stakeholders possible solutions/ideas or advances in the selected topic through considering local practices (further research or administrative action).
		Complete the research report by Week 5 and submit or send the manuscript for publishing in an appropriate newsletter.
2	Objective 8: Incorporate drug information services to pharmacotherapy services by proactive information.	
	Action Plans	Identify drug-related needs or potential drug-related problems in the patients for whom the queries were answered.
		Collect further information specific to the patient considering subjective and objective information.
		Assess medication-related risk for harm in patients with available information.
		Communicate with the healthcare provider on how to improve pharmacotherapy. Communicate directly with the patient, if necessary.
		Provide/use proper documentation.
3	Objective 9: Perform critical appraisal of a selected therapy topic and give a short (15 min) oral presentation. (See Appendix 2 for a rubric to assess the educational presentation).	
	Action Plans	Select a therapy topic to review.
		Prepare the presentation slides and a handout for this topic.
		Give the educational presentation to a group of fellow students or colleagues and the preceptor.

No additional objectives for week 4; Repeat Objectives 4–6 of Week 2.

TABLE 13.4 Learning Objectives for Week 5

No	Topic: Drug Information Services and Newsletter Communications	
1	Objectives 4–6 continue in Week 5	
2	Objective 10: Author, or at least review, a safety-focused drug therapy monograph that will be submitted to the pharmacy and therapeutics committee for a drug to be considered for addition the formulary.	
	Action Plans	Work with a practitioner that wants to add a drug to the formulary.
		Prepare appropriate monograph using the P&T committee's standard format.
		Identify the primary literature and manufacturer's information that will be necessary to complete the monograph.
		Use knowledge skills to organize the basic information necessary for the P&T committee to assess the efficacy and safety of this proposed addition to the formulary.
		Carefully document all the sources used to prepare the monograph.
3	Objective 11: Prepare the training report.	
	Action Plans	Compile all documents relevant to each task completed.
		Prepare a self-reflection report of the training.
		Revise the report based on feedback.
		Review knowledge, skills, and competencies gained during the training.
		Undergo end of training evaluation by the preceptor.

CONCLUSION

Drug information training is an essential component of pharmacy education. Pharmacists are expected to provide evidence-based recommendations regarding rational drug use to patients and healthcare professionals on a daily basis. Systematic experiential training of drug information skills focuses on search, appraisal, and writing skills. Pharmacists, irrespective of the practice or research setting, need to learn and use strong drug information skills.